

Sean P. Florez

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Research Profile

Computational materials scientist building **AI-accelerated, autonomous workflows** for simulation and interfacial materials, spanning **large-scale MD/DFT campaigns** (LAMMPS, VASP, Quantum ESPRESSO), **ML models for quantum chemistry and experimental data**, **force fields calibrated against experiment**, and **HPC orchestration** across DoD, DOE, and academic clusters. Interested in representation learning and agentic systems that close the loop between **simulation, experiment, and scientific decision-making**.

Skills

Programming and ML: Python; scikit-learn, TensorFlow, imbalanced-learn; RDKit, Mordred, SOAP descriptors; NumPy, Pandas, SciPy, Matplotlib; pytest, CI, package and CLI design.

Simulation: LAMMPS, NAMD, VASP/VASPsol, Quantum ESPRESSO, Materials Studio; INTERFACE FF, CHARMM, CVFF, PCFF, COMB3, EAM; model building, parameterization, experimental validation.

HPC and systems: Slurm and PBS Pro, job arrays, checkpoint-restart; ALCF, CU Alpine, DoD HPC; Linux, git, systemd, SQLite, Docker, GitHub Actions, MCP; data lineage and evidence-bundle post-processing.

Research Experience (Materials & AI/ML)

Graduate Research Assistant

University of Colorado Boulder, Oct 2024–Present

Heinz Interfaces Laboratory, Prof. Hendrik Heinz

- Ran the production shear campaign for OH-, F-, and mixed-terminated Ti_3C_2 MXene bilayers, running **780 NVT trajectories** on CU's Alpine cluster, zero failures, against a protocol frozen before production and audited read-only afterward. Termination sets interlayer shear resistance across a threefold range ($\tau_{0.5}$ 42–135 MPa).
- Designed and ran the MD campaign behind a lead-author manuscript on N-ethylcarbazole dehydrogenation over Pt nanocrystals; reactive configurations concentrate at low-coordination sites (vertices carry $5.4\times$ the per-atom contact density of {100} terraces). Wrote the technical narrative for the ALCF Director's Discretionary award funding the next campaign (**$\sim 26,150$ node-hours**).
- Parameterized INTERFACE FF models for copper oxides as the first-year deliverable of an **industry-funded force-field program**, calibrated exclusively against experiment. Validated a Cu_2O parameter family within $\pm 5\%$ of both experimental elastic references and verified **52 configurations** for LAMMPS, msi2lmp, CHARMM, and NAMD.
- Built a SOAP-embedding retrieval index over **3,009 structures and 6.7 million atoms** to assign starting atom types and parameters from nearest structural neighbors, with confidence thresholds and explicit coverage gaps.

High-Performance Computing Intern

Air Force Research Laboratory, May–Aug 2025

Materials & Manufacturing Directorate, Wright-Patterson AFB

- Built LAMMPS shear workflows for hydrated Ti_3C_2 MXene bilayers and ran parameter sweeps on **DoD HPC**.
- Fit slip probability $p(\tau)$ against shear stress with logistic models, revealing a non-monotonic hydration effect. Dry bilayers hold to 103 MPa, a quarter monolayer drops them to 8 MPa, and full coverage recovers to 39 MPa.
- Delivered the first MD evidence that hydration tunes MXene interlayer shear strength by an order of magnitude.

Research Software & Infrastructure

OpenSDL, computational and autonomous laboratory framework

2026–Present

github.com/fi-sean03/OpenSDL | seanflorez.com/OpenSDL

Apache-2.0; $\sim 26\text{k}$ non-test lines of Python, ~ 600 test functions, CI

- Declare what a laboratory can do, execute those declarations as reproducible workflows, preserve the evidence, and feed each result back into the choice of the next experiment.
- Durable DAG runtime with retries, timeouts, resource leases, restart reconciliation, and policy checks over SQLite metadata and content-addressed artifact storage; adapters for a simulated lab, local compute, and human tasks, with Slurm and hardware integration in progress.
- Implemented campaign scoring and next-round selection. A reference three-dye formulation search matched an unseen target recipe within **0.5 percentage points**, evaluating **96 candidates per round**.

agent-fleet, persistent research automation

2026–Present

github.com/fi-sean03/agent-fleet

- Built a research harness based on a **15-agent fleet** operating for months, with isolated workspaces, persistent conversations, and inter-agent messaging.
- Ran multi-day computational campaigns with recovery after reboots, credential expiry, and preemption.

HPC & Cloud Throughput Infrastructure

2025–Present

- Developed **subjob** to multiplex short tasks in one HPC allocation, with shared worker pools and heartbeat recovery.
- Built **CloudComputeManager** for GPU spot-instance provisioning, environment setup, pipeline execution, pre-emption recovery, and per-run cost tracking; used it for MXene shear sweeps.

Selected AI/ML for Materials Projects

QM9 Quantum-Chemical Property Prediction (TensorFlow FNNs)

GitHub: [qm9-smiles-predictor](#)

- Four reproducible pipelines (single-task, multitask, autoencoder, hybrid Mordred+ECFP4) predicting nine quantum-chemical properties for ~130k molecules.
- Reproduced and critiqued SMILES-FNN baselines; showed where multitask learning underperforms and where hybrid descriptors reduce MAE relative to Mordred-only models, for dipole moment and polarizability in particular.

Automated Phase Identification in Experimental pXRD (opXRD)

GitHub: [opxrd-ml-binary-phase](#)

- Ingested the ~92k-pattern opXRD archive (~2.4% labeled) and formulated binary classification for the rare As₈O₁₂ phase, on an interpolated 2θ grid with normalized intensities.
- Trained logistic regression and random forest classifiers; applied SMOTE against extreme class imbalance and reached a 0.80 minority-phase F1 with balanced precision and recall on held-out data.

Optimization of VASPsol Solvation Parameters (DFT + Surrogate)

Hennig Group, University of Florida

- Benchmarked VASPsol against experimental energies from the [Minnesota Solvation Database](#) for neutral and charged solutes in water; optimized cavity and screening parameters by grid search and Nelder–Mead to reduce RMSE, and outlined a surrogate-model strategy replacing high-count parameter sweeps.

Additional Experience

Engineering Contractor, Rocky Mountain Scientific Laboratory

Jun–Aug 2026

Quality system for a client affiliate's new energetic-materials production line

- Developed lot-release records covering inspections, approval authority, hold criteria, and nonconformance disposition, with item-level traceability to the governing military specification. Built a controlled, single-source document set with automated link, terminology, and source-consistency checks, and a program tracker that generates the Kanban board, decision log, and requirements traceability matrix.

S&T Scouting Intern

The Joint Staff, J7 / Future Technology Office, Aug–Oct 2025

- Produced **eight technical briefs** on propulsion and energy concepts, assessing readiness, risk, and experimentation paths for TRL 3–5 technologies across Service laboratories and DoD R&D organizations.

Technology Commercialization Intern

Idaho National Laboratory (DOE), May–Aug 2024

- Conducted **20 Energy I-Corps interviews**, developed a commercialization plan for a laboratory technology, built AI and natural-language tooling for the technology-transfer team, and supported a DOE proposal awarded **\$2M**.

Materials Modeling Intern

Air Force Research Laboratory, May–Dec 2023

- Automated high-throughput DFT screening of low-work-function emitters (Quantum ESPRESSO, ASE, pymatgen).

Education

University of Colorado Boulder

Expected May 2028

Ph.D., Materials Science & Engineering. Advisor: Prof. Hendrik Heinz.

University of Florida

May 2024

B.S., Materials Science & Engineering. Minor in Chemistry.

Publications & Outputs

Co-author: *INTERFACE Force Field for Alumina with Validated Bulk Phases and a pH-Resolved Surface Model Database for Electrolyte and Organic Interfaces*. [arXiv:2601.12570](#), 2026 (preprint).

Lead-author manuscripts in preparation: *Modeling of MXene Shear Properties via Atomistic Molecular Dynamics Simulations; Understanding Hydrogen Release Mechanisms of N-Ethylcarbazole on Pt Nano-Crystal Catalysts*.

Open-source releases: [MolSAIC](#) (MD system builder), [pdb2msi](#) (PDB-to-Materials Studio converter).

Leadership

Co-Founder, [LabLink Initiative](#) (STEM career access)

Aug 2024–Present

Partnerships with three Sacramento-area community colleges to move students into DOE Office of Science laboratory internships (CCI and SULI), plus a Phi Theta Kappa consortium webinar that drew more than 250 students.